

10 BI Prompts

Task	Prompt	AI Tool Used	Output (Summary)	Evaluation
1. Data Cleaning	<i>I have an Indian Crop Production dataset containing State, District, Crop, Season, Year, Area, Production, and Yield. Suggest a step-by-step data cleaning process before building a Business Intelligence dashboard.</i>	ChatGPT	AI suggested checking missing values, removing duplicates, validating data types, handling outliers, and standardizing categorical values.	The response was well-structured and covered all essential preprocessing steps. It was useful for preparing the dataset before visualization.
2. Define KPIs	<i>Suggest three important KPIs for analyzing Indian crop production using a Business Intelligence dashboard.</i>	ChatGPT	AI recommended Total Production, Average Yield, and Number of Crops, along with explanations of why each KPI is useful.	The KPIs matched the project objectives and were directly applicable to the dashboard.
3. Executive Summary	<i>Write a 150-word executive summary for a dashboard showing crop production trends, top-producing crops, seasonal production, and regional analysis.</i>	ChatGPT	AI produced a concise summary highlighting major production trends and business insights.	The summary was clear but required minor editing to ensure it reflected the actual dashboard findings.
4. SQL Query Generation	<i>Generate an SQL query to calculate the total crop production for each state in descending order.</i>	ChatGPT	AI generated a SQL query using GROUP BY , SUM() , and ORDER BY to	The query was syntactically correct and suitable for relational database analysis.

			rank states by total production.	
5. Chart Recommendation	<i>Which chart is best for comparing crop production across different crops, and why?</i>	ChatGPT	AI recommended a horizontal bar chart because it clearly compares values across multiple categories.	The recommendation matched BI visualization best practices and improved dashboard readability.
6. Correlation Interpretation	<i>Explain the possible relationship between cultivated area and crop production in simple business language.</i>	ChatGPT	AI explained that larger cultivated areas often result in higher production but noted that yield and farming practices also influence output.	The explanation was balanced and avoided making unsupported assumptions.
7. Data Quality Issues	<i>Identify possible data quality issues that should be checked in an agricultural production dataset.</i>	ChatGPT	AI listed duplicate records, missing values, inconsistent spellings, invalid numerical values, and incorrect date formats.	The response provided a comprehensive checklist that was useful during data validation.
8. Slide Headline	<i>Write a professional presentation headline for a chart showing the top 10 crops by production.</i>	ChatGPT	AI suggested the headline: " Crop Production is Concentrated Among a Few Major Crops. "	The headline was concise, business-oriented, and suitable for an executive presentation.

9. Business Recommendation	<i>Based on high production concentration in a few crops, provide three business recommendations for policymakers.</i>	ChatGPT	AI recommended supporting crop diversification, improving irrigation, and using BI dashboards for continuous monitoring.	The recommendations were practical and aligned with the project's findings.
10. Explain a Technical Concept	<i>Explain Business Intelligence dashboards to a non-technical audience using simple language.</i>	ChatGPT	AI described BI dashboards as interactive tools that transform raw data into easy-to-understand charts and insights for better decision-making.	The explanation was simple, accurate, and appropriate for non-technical stakeholders.

3 Prompt Variations

BI Task Chosen

Writing an Executive Summary from a Business Intelligence Dashboard

Variation 1 – Minimal Prompt

Prompt

Write an executive summary of this crop production dashboard.

Output (Summary)

The AI generated a short and generic summary mentioning crop production trends and dashboard insights. However, it lacked specific details, business context, and actionable recommendations.

Variation 2 – Context-Rich Prompt

Prompt

The dashboard is based on the Indian Crop Production Dataset (1997–2015). It includes KPI cards for Total Production, Average Yield, Number of Crops, and Total Area, along with charts showing production trends, top-performing crops, and seasonal production. Write a 150-word executive summary for agricultural policymakers.

Output (Summary)

The AI produced a more detailed summary highlighting production trends, major crops, and the importance of data-driven agricultural planning. The content was relevant to the dataset and suitable for a policymaker audience.

Variation 3 – Structured + Role Prompt

Prompt

Act as a Senior Business Intelligence Analyst. Using the Indian Crop Production Dashboard, write a 150-word executive summary for a government board. Structure the response into three sections: Key Findings, Business Impact, and Recommendations. Use professional language, avoid technical jargon, and provide only evidence-based observations without making unsupported assumptions.

Output (Summary)

The AI generated a well-structured executive summary with clear sections, focused business insights, and practical recommendations. The response was concise, professional, and aligned with the dashboard findings while avoiding unsupported conclusions.

Comparison

The **Structured + Role Prompt (Variation 3)** produced the best output because it clearly defined the AI's role, target audience, response structure, and writing constraints. Unlike the minimal prompt, which generated a generic summary, and the context-rich prompt, which improved relevance but lacked consistent organization, the structured prompt delivered a

professional, business-focused summary with actionable recommendations. The biggest improvement came from combining **role prompting**, **context**, **explicit formatting instructions**, and **constraints**, which significantly enhanced the clarity, accuracy, and usefulness of the AI-generated response.